

Zee Pearson

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SUMMARY

Embedded Software Engineering student with hands-on experience developing microcontroller-based sensing, data-logging, automation, and robotics systems using C/C++, Java, ESP32-C6, Raspberry Pi Pico 2 W, UART, Zigbee, and OpenCV. Seeking an embedded software or firmware opportunity to contribute strong problem-solving, hardware-software integration, and debugging skills while expanding technical expertise through real-world product development.

WORK EXPERIENCE

Embedded Systems Research Intern

June 1 - August 7, 2026

Blue Devil Adventure - Lawrence Technological University

- Programmed an ESP32-C6 to poll Third Reality Zigbee moisture sensors and acquire environmental readings.
- Designed a UART link between the ESP32-C6 and Raspberry Pi Pico 2 W for dependable sensor-data transfer.
- Developed Pico 2 W firmware for microSD logging and debugged the complete sensor-to-storage embedded pipeline.
- Built REST API endpoints and a web dashboard for live and historical readings plus remote pump and valve control.

Caddie

April 2018 - Present

Bloomfield Hills Country Club - Bloomfield Hills, MI

- Provide yardage, club-selection, shot-strategy, and course-management guidance while delivering dependable client service.

EDUCATION

Bachelor of Science in Embedded Software Engineering

Expected Graduation: Fall 2027

Lawrence Technological University - Southfield, MI

- Current GPA: 3.7 | Relevant coursework: C++, VHDL/HDL Circuit Design, MATLAB, Real-Time Systems

Associate of Applied Science in Software Engineering

Aug 2022 - May 2024

Oakland Community College - Bloomfield Hills, MI

- GPA: 3.8, Magna Cum Laude | Java, Python, HTML/CSS, Database Management, Data Structures

ORGANIZATIONS/PROJECTS

Vice President

Fall 2025 - Present

Institute of Electrical and Electronics Engineers (IEEE) - LTU Student Branch

- Organized research-focused technical events that connected students with engineering topics, opportunities, and professional development resources.
- Expanded student participation by recruiting 20 new members through outreach, event promotion, and direct engagement.
- Led planning and event operations for programs serving more than 50 attendees, coordinating logistics and supporting successful execution.

Smart Home Automation System - Team Project

January - April 2026

Lawrence Technological University

- Designed a Tiva C smart-home control system and implemented UART communication between embedded hardware and a mobile application.
- Developed feedback-based control logic to regulate temperature dynamically from user input.

Robotics Team Member - Da Bears 247 * The Crew 7769

September 2019 - May 2024

FIRST Robotics

- Programmed autonomous robots using Java and FRC libraries.
- Worked with PWM ports to control motors, servos, and actuators.
- Developed autonomous robot behavior using libraries such as Pathfinder for path planning and trajectory control.
- Built a computer-vision-guided pickup system using OpenCV to detect game pieces and deliver them to another location autonomously.

AWARDS/CERTIFICATIONS

- Information Technology Specialist: JavaScript (Jan 2023), Java (Apr 2023), HTML/CSS (Mar 2023) - Certiport
- Dean's List: Fall 2024, Spring 2025, Fall 2025, and Spring 2026
- Maintained a 3.50+ GPA every semester at LTU while simultaneously serving as IEEE Vice President.
- Member of Kappa Phi Honor Society